

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Ammeter in series (1) With thermistor (1) Voltmeter in parallel with thermistor (1) CORRECT SYMBOLS ONLY	3			3		3
	(b)	(i)		5 points plotted correctly $\pm < 1$ small square (2) 4 points plotted correctly $\pm < 1$ small square (1) 3 or less points plotted correctly $\pm < 1$ small square (0) Smooth curve $\pm < 1$ small square (1) Don't accept wispy, double, disjointed, dot to dot curves		3		3		3
		(ii)		Decrease (1) at decreasing rate (1)		2		2		2
	(c)	(i)		From candidate's graph $\pm < 1$ small square expect 3 600 – 3 900 [Ω]		1		1		1
		(ii)		Selection of: $\text{current} = \frac{\text{voltage}}{\text{resistance}}$ (1) Substitution (1) $\frac{12}{\text{answer from (i)}}$ = correct answer (1) If resistance 3 600 [Ω] then 0.0033 [A] 3 700 [Ω] then 0.0032 [A] 3 800 [Ω] then 0.0032 [A] 3 900 [Ω] then 0.0031 [A] 4 000 [Ω] then 0.0030 [A] Accept answer in mA if correct and unit changed If no workings shown answers of 3.3 [A] etc award 2 marks only	1 1	 1		3	2	
	(d)			Resistance changes by 5 400 – 1 400 (1) = 4 000 [Ω] (1) so it is suitable Award 2 marks only if the conclusion is present			2	2		
				Question 6 total	5	7	2	14	2	9